SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

B. Tech. Project Poster AY 2026-27

TITLE: ForgeFlow: An AI-Native End-to-End Software Development Lifecycle Automation Platform

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Celebrating 50 Years of Excellence

INTRODUCTION

An LLM can now generate a web app in minutes. Knowing that the app works, and knowing when to stop fixing it, is still hard.

- Asking a model to keep improving its own code can make the code less secure. (Shukla et al., 2025)
- Fewer than 1 in 3 model-generated infrastructure deployments succeed on the first attempt. (Zhang et al., 2026)
- Most of a repair loop's gain arrives in the first 1–3 rounds; later rounds mostly spend tokens. (Kiecker et al., 2026; Arimbur, 2026)
- Developers steer coding agents closely rather than hand over control. (Huang et al., 2025; Tang et al., 2026)

The gap

	Research repair agents	LLM test / repair tools	Prompt-to-app builders	ForgeFlow
Repair with an iteration cap and stall detection	×	×	×	✓
Hands back to a person when repair stalls	×	×	×	✓
Requirements compiled into traceable tests	×	*partial*	×	✓
Automated cloud deployment	×	×	✓	✓
Re-tests the deployed, live URL	×	×	×	✓
Any stage can be skipped or revisited	×	×	*partial*	✓

Capabilities as described in the cited work, adapted from Table I of our paper; not a performance comparison.

ForgeFlow takes a web-app idea, given as screenshots or a structured spec, to a live URL that has been tested, and lets a person enter, refine or skip any stage on the way.

OBJECTIVES & AIMS

01 Non-linear control. No stage is mandatory; every artifact is versioned and can be revisited.

02 Generation. Screenshots or a spec become a full-stack app on one fixed stack.

03 Requirements to tests. Acceptance criteria compile into unit and E2E tests, each traceable to its criterion.

04 Bounded self-healing repair. A diff-aware loop with caps, stall detection and escalation to a person. (Core contribution)

05 Infrastructure reasoning. Work out what each app needs, then deploy it.

06 Live validation. Re-test the deployed URL; repair and redeploy on failure.

07 Isolation. Generated code runs only in a per-project sandbox.

08 Measurement. First-pass vs. post-repair success, cost and latency, reproducibly.

Hypotheses under test

H1 Bounded repair captures most of the achievable gain at a fraction of an unbounded loop's cost.

H2 A minimal, diff-aware context repairs more per token than broad repository context.

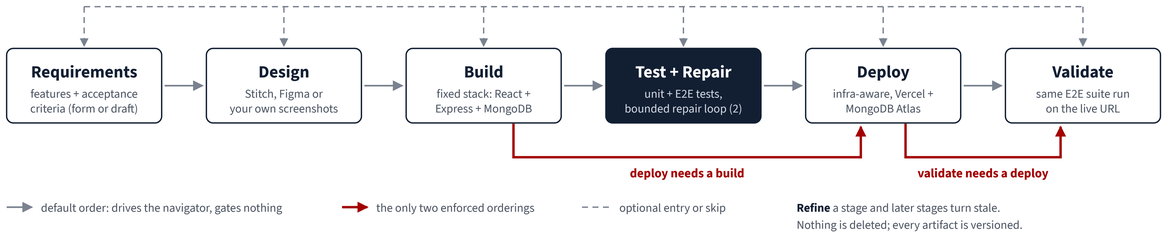
H3 Re-testing the live URL catches faults that sandbox tests miss.

METHODOLOGY

1 Six stages, none mandatory

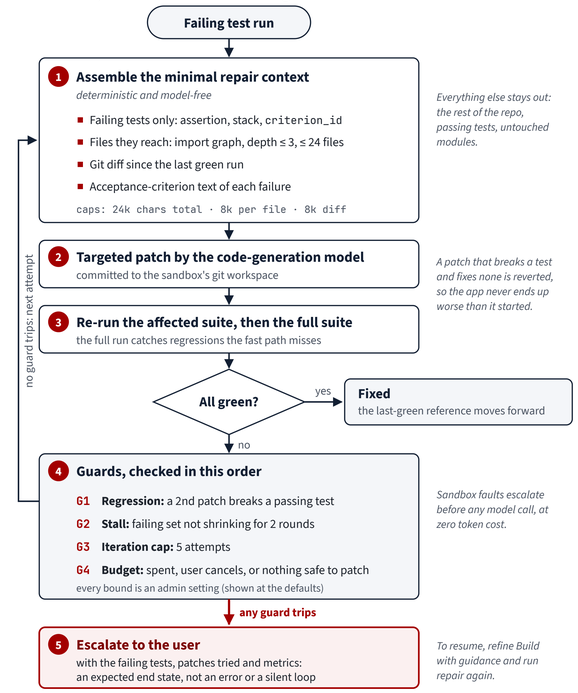
The conductor, not the UI, decides which transitions are legal.

You can enter at any stage, skip any stage, or jump back to refine one.

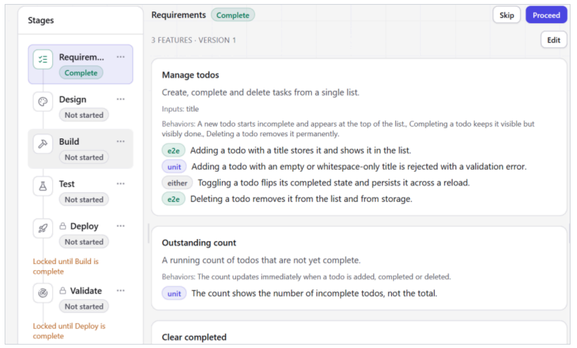


Refine a stage and later stages turn stale. Nothing is deleted; every artifact is versioned.

2 Bounded, diff-aware self-healing repair

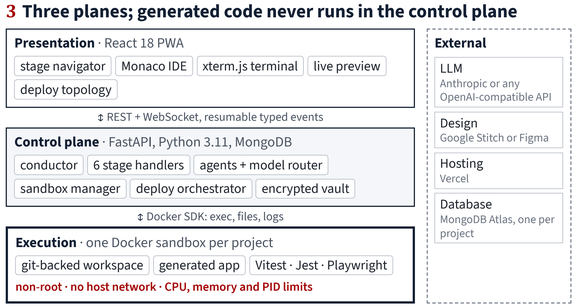


Why a diff: generated tests lose validity when code changes, and highlighting the change is the recommended fix (Haroon et al., 2026). **Why a cap:** most gain comes in the first rounds (Kiecker et al., 2026).



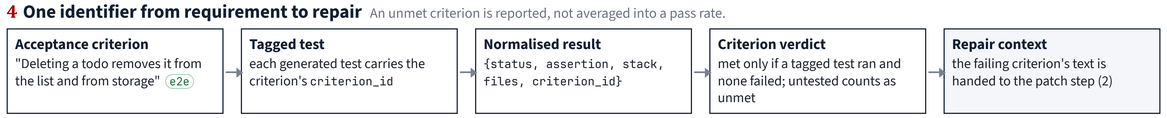
The workspace (real UI). Each acceptance criterion carries a `unit` or `e2e` tag. Deploy and Validate stay locked until Build and Deploy finish; every other stage can be skipped.

3 Three planes; generated code never runs in the control plane

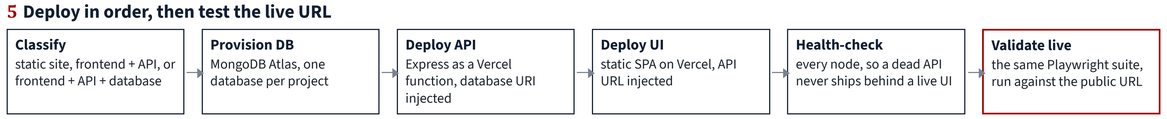


4 One identifier from requirement to repair

An unmet criterion is reported, not averaged into a pass rate.



5 Deploy in order, then test the live URL

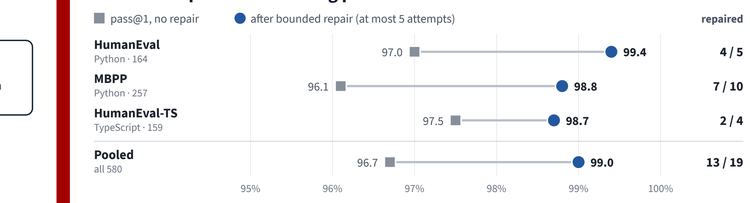


Live failure → same bounded loop → redeploy → re-validate (at most 2 cycles). Tokens are platform-owned or the user's own, Fernet-encrypted at rest and never sent to the browser.

RESULTS & CONCLUSIONS

One model (GLM-5.3-flash) through ForgeFlow's own model interface; one reproducible run, 2 Oct 2026. Problems from HumanEval (Chen et al., 2021), MBPP (Austin et al., 2021) and HumanEval-TS (Cassano et al., 2023).

Bounded repair on 580 coding problems



Per-benchmark 95% Wilson intervals overlap. The pooled gain is significant (exact McNemar p = 0.00024); per benchmark only MBPP is (p = 0.016).

What happened to the 19 first-attempt failures



+5.7% model calls for the repairs (33 on top of 580)

0 runs reached the cap of 5; the stall guard did the bounding

Whole applications

2 of 10 specs run, once each



Every unmet criterion was a browser test. Browser tests import no app source, so the minimal context had nothing to edit and the loop escalated (reason: `blocked`) in about 1.3 s instead of guessing. Neither app was delivered.

Conclusions

- **Repair pays where a failing test reaches the code:** 13 of 19 first-attempt failures fixed for 5.7% more model calls, and the guards, not the cap, ended every unproductive run.
- **It knows when to stop:** all 580 problems and both apps ended with a recorded, typed reason, and each unmet criterion traces to a failing test.

H1 SUPPORTED at function level; one model, one run

H2 NOT YET TESTED context-ablated arm still to run

H3 QUALITATIVE e.g. a database URI that failed only once deployed

Next: map failing browser tests to the pages and routes they visit, so repair can act on them; then run all 10 apps.

REFERENCES (APA)

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OUTCOME OF THE PROJECT WORK

IEEE conference paper
Submitted; under review.

Invention disclosure
Bounded diff-aware self-healing delivery; submitted for filing.

IBM Bob 2.0 Hackathon
ForgeFlow submitted, lablab.ai, 25–27 Sep 2026.

Working platform
65 phases, about 54k lines, 2,323 automated tests; all 2,253 run on the dev host pass.

Platform CI/CD
Six-job CI; health-checked deploys. Rollback drill: 807 requests, 0 failed.

TRY IT LIVE

Scan to open the deployed platform.
`forgeflow-ecru.vercel.app`
No industry collaborator on this project.

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